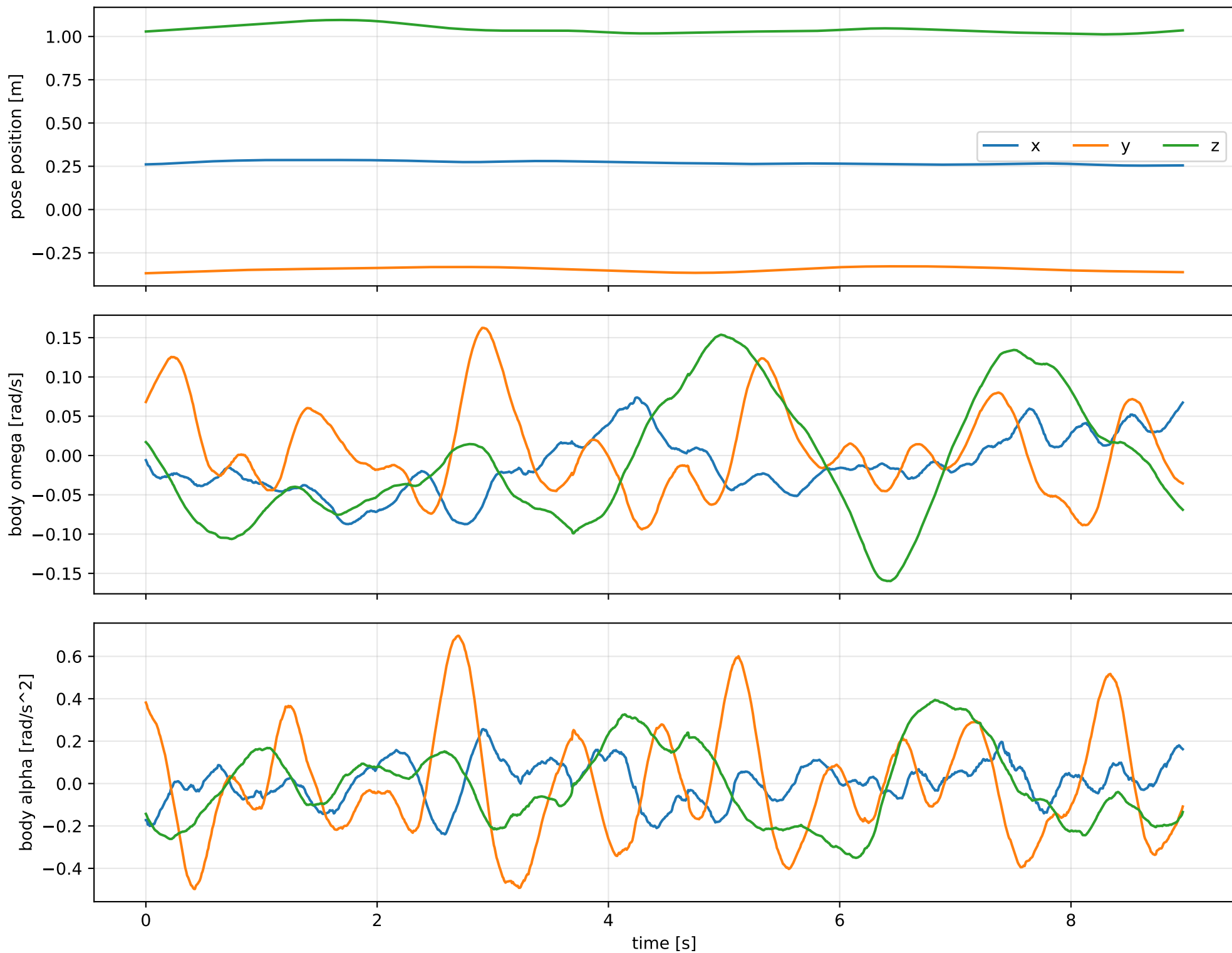
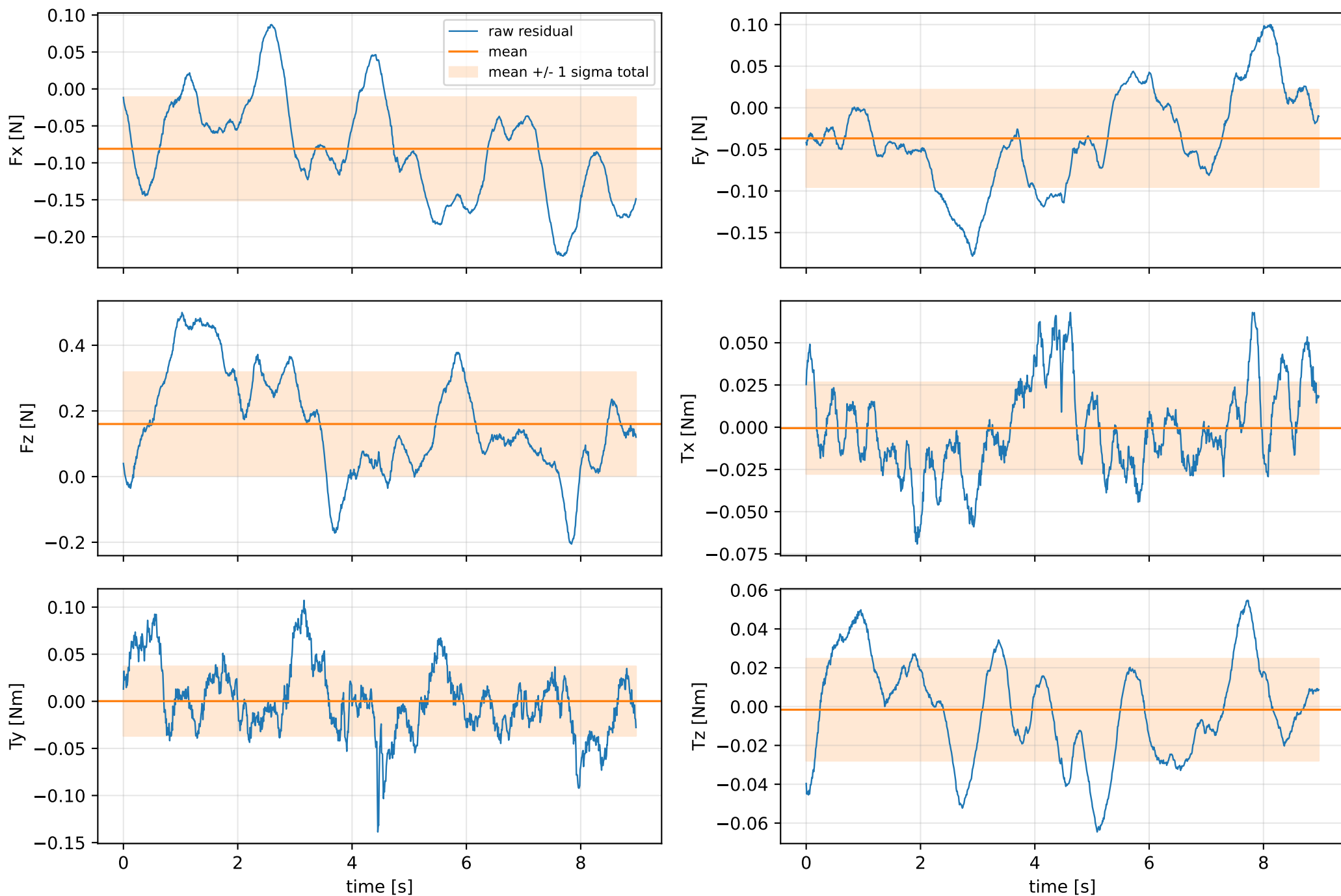


inertia_over_mass_all_nominal: SG trajectory and derived motion



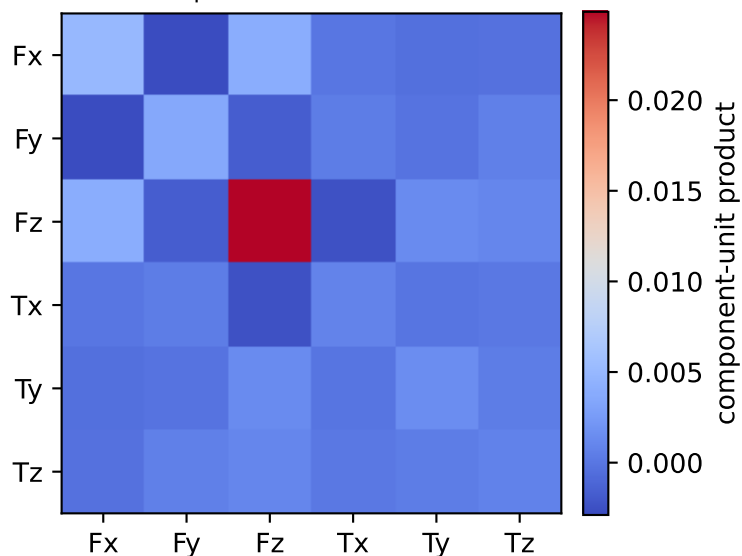
inertia_over_mass_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



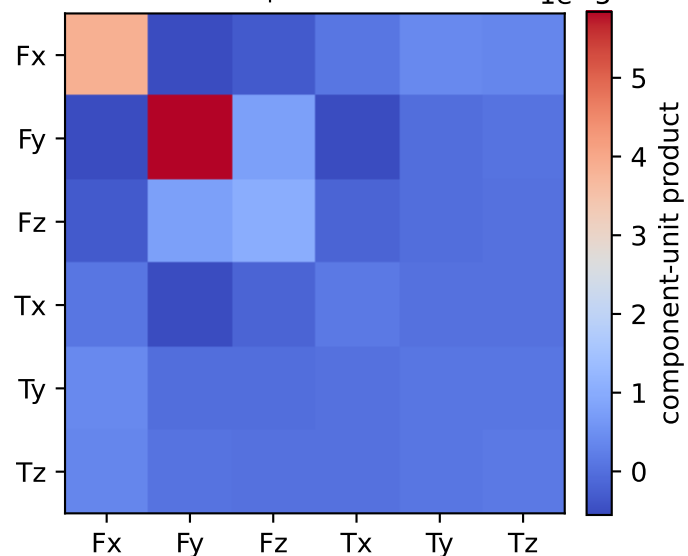
vector RMS about zero: force 0.25831 N, torque 0.052783 Nm; component std about mean: force [0.0699 0.0585 0.1577], torque [0.0271 0.0369 0.0263]

inertia_over_mass_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0. 0. 0.001 0.002 0.006 0.026]

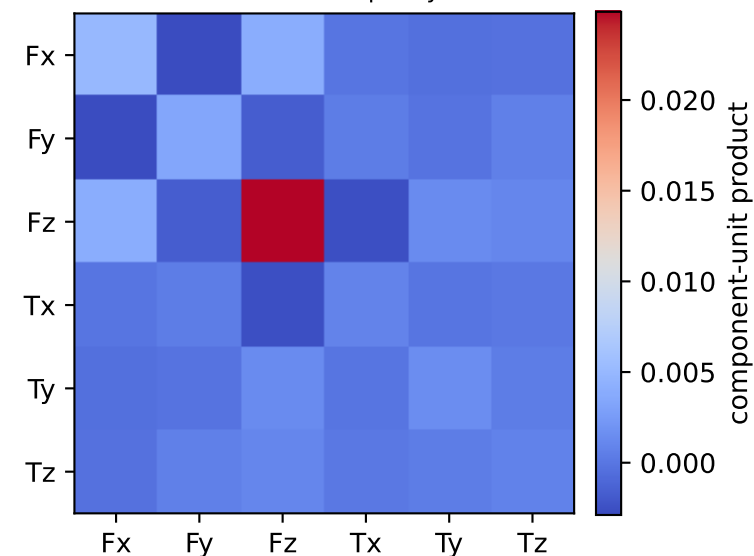
empirical total covariance



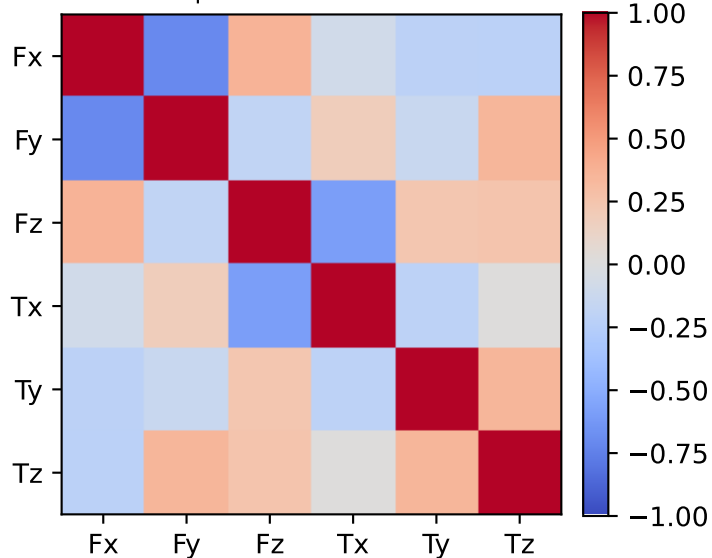
mean full-SG prediction covariance $1e-5$



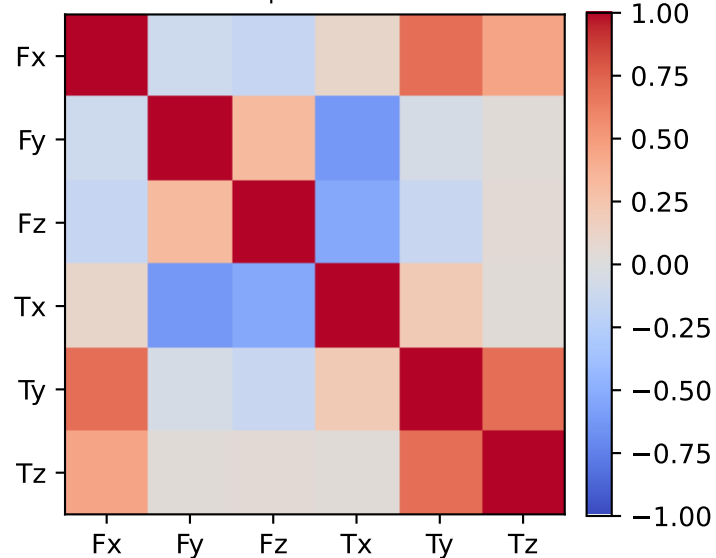
PSD excess model discrepancy covariance



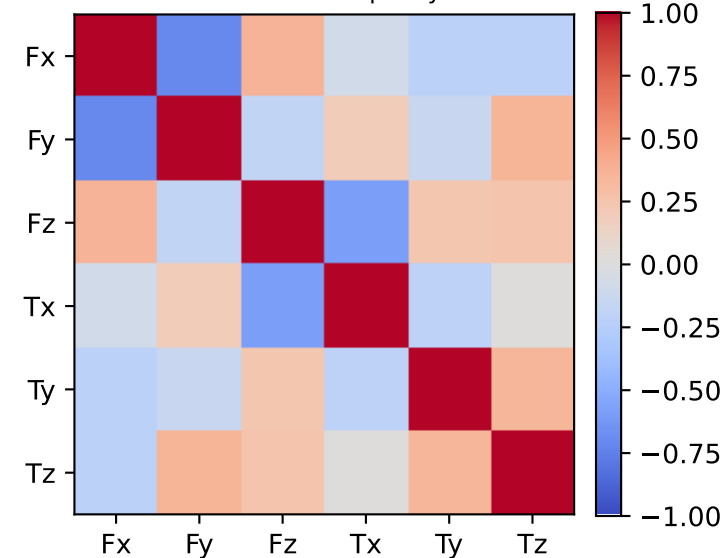
empirical total correlation



mean full-SG prediction correlation

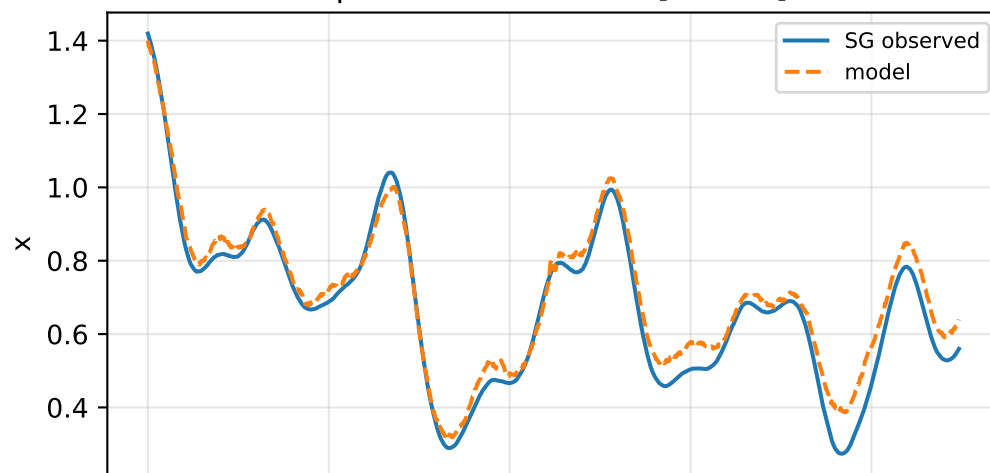


PSD excess model discrepancy correlation

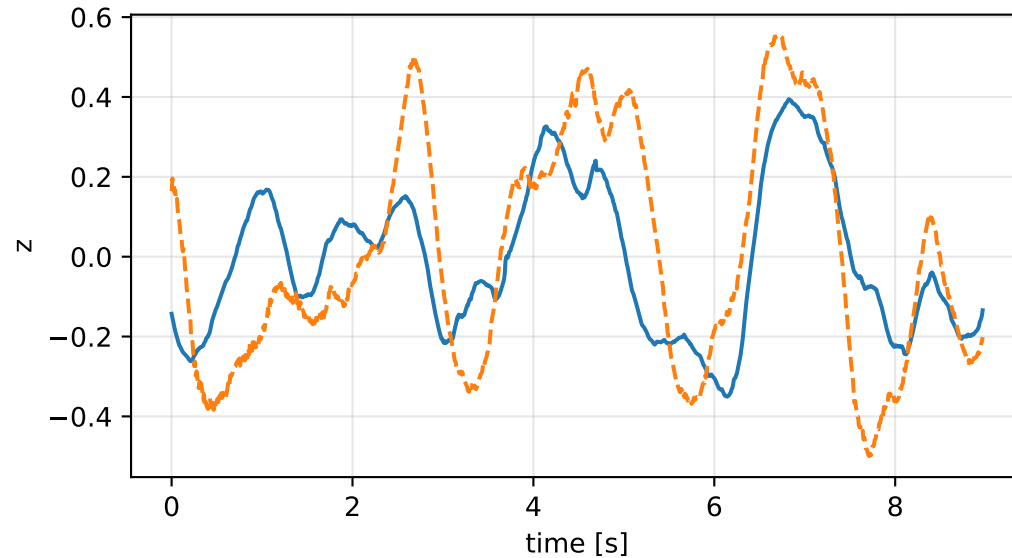
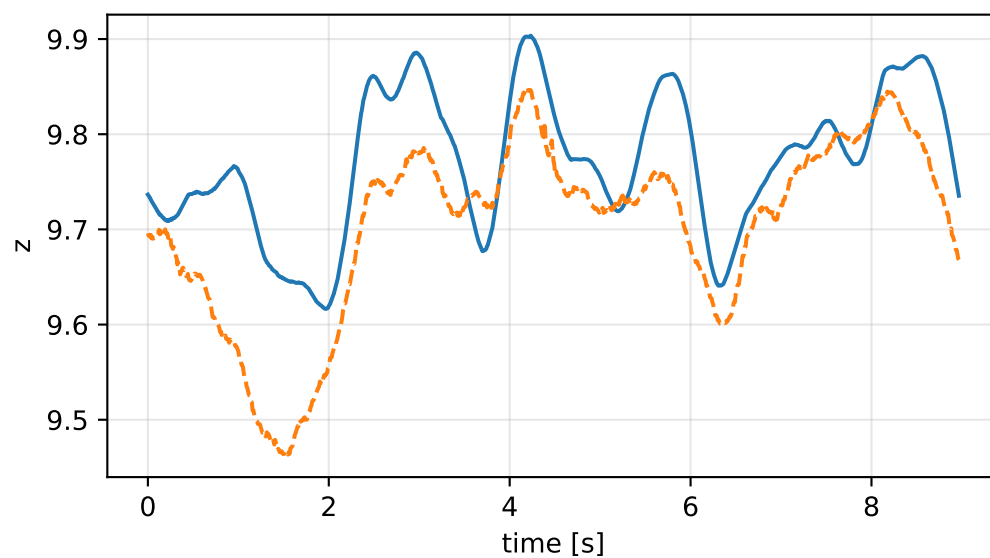
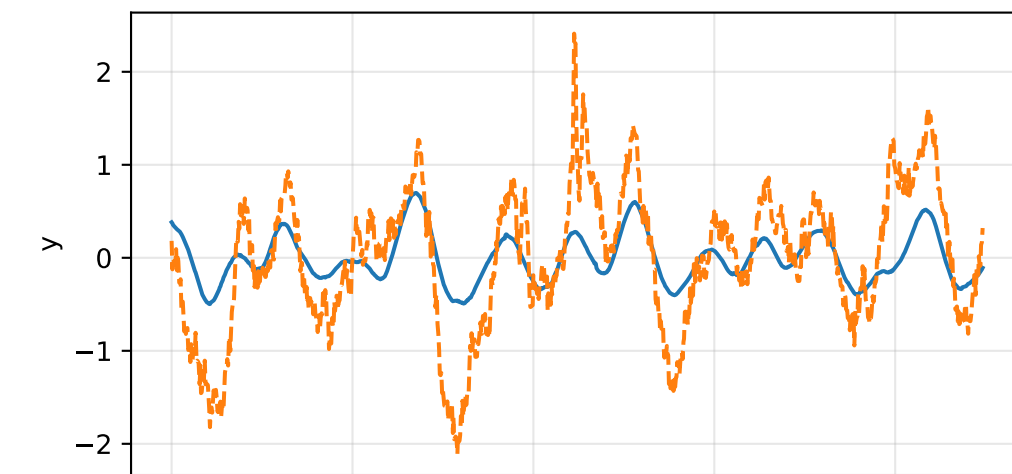
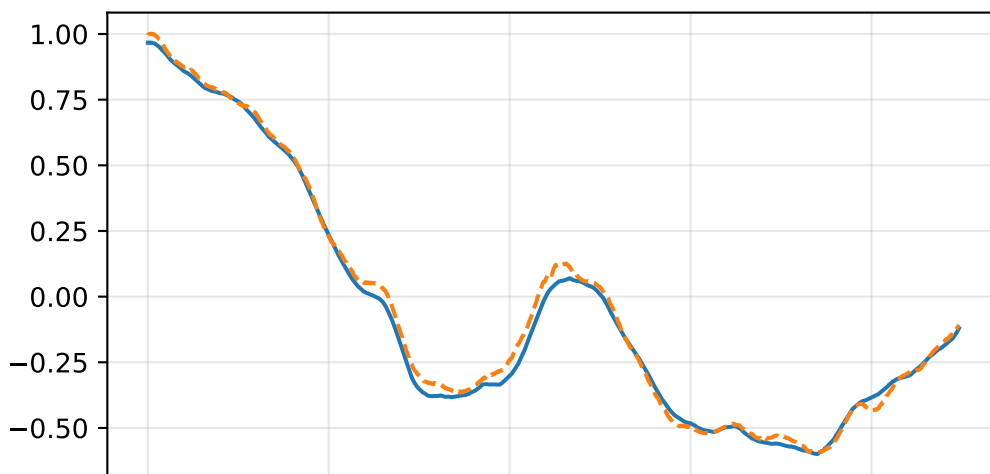
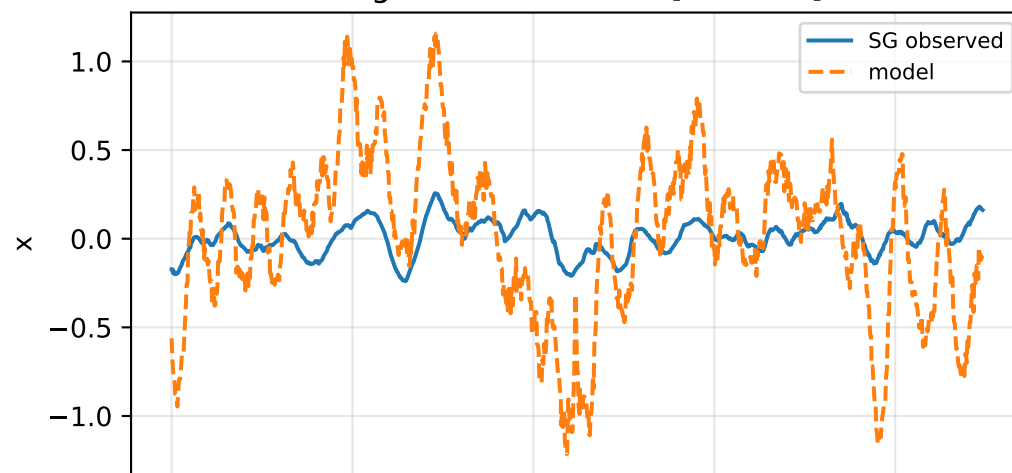


inertia_over_mass_all_nominal: acceleration residual objective

specific acceleration [m/s^2]

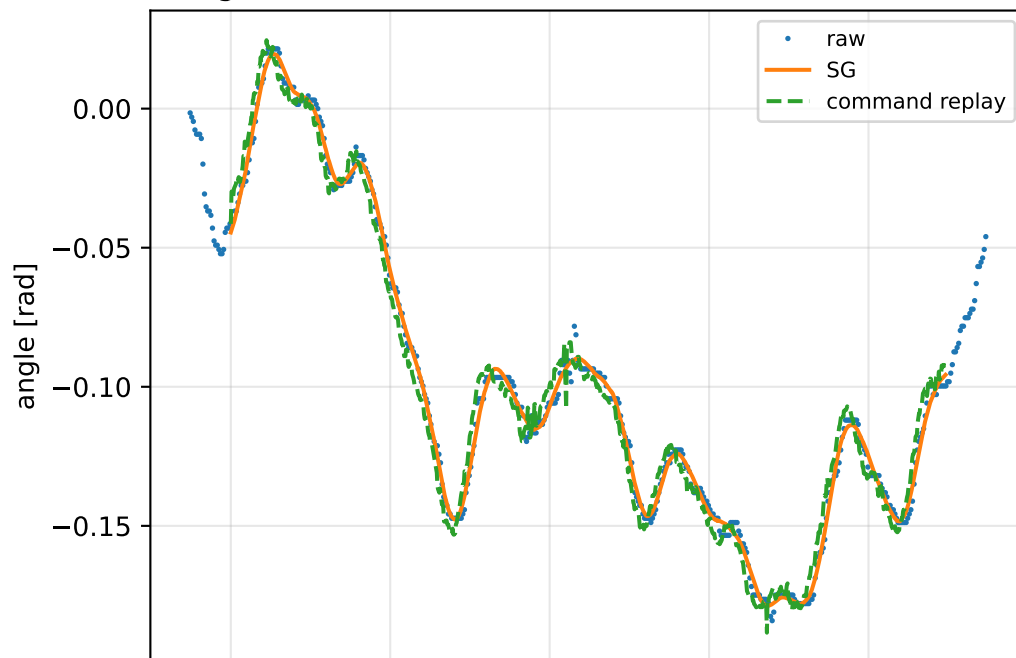


angular acceleration [rad/s^2]

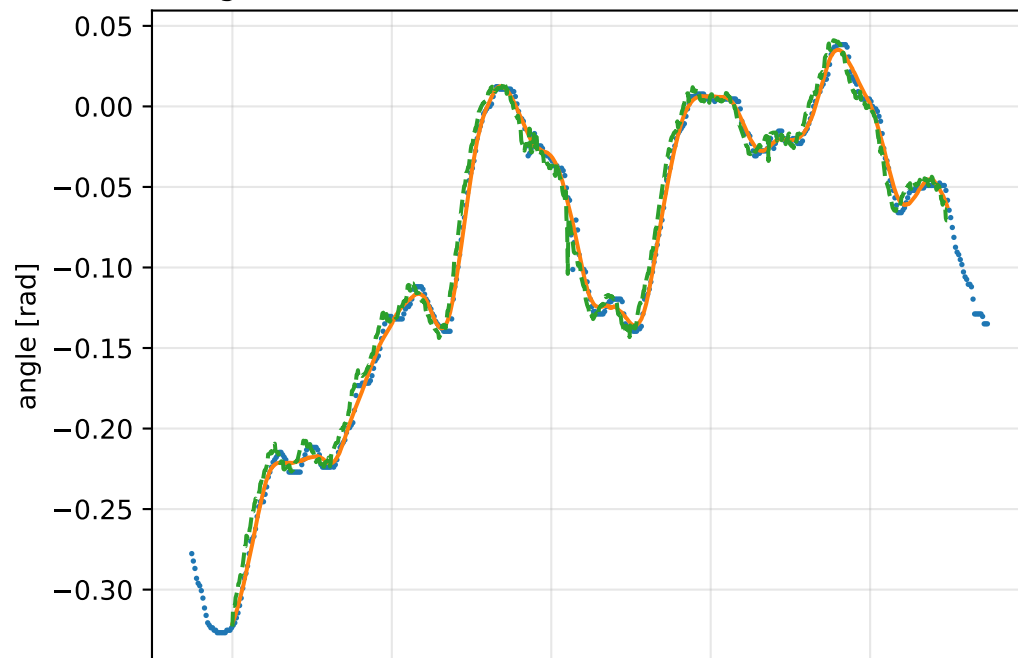


inertia_over_mass_all_nominal: gimbal measurement audit (objective=measured_sg)

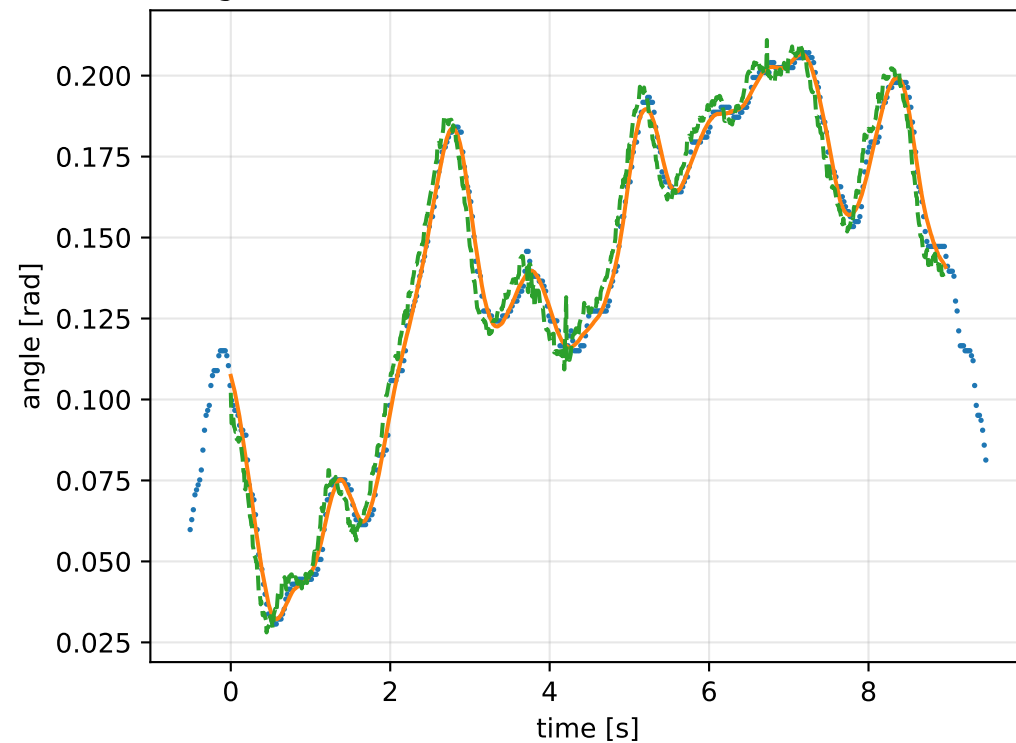
gimbal 1: RMSE=0.006345, max=0.01494 rad



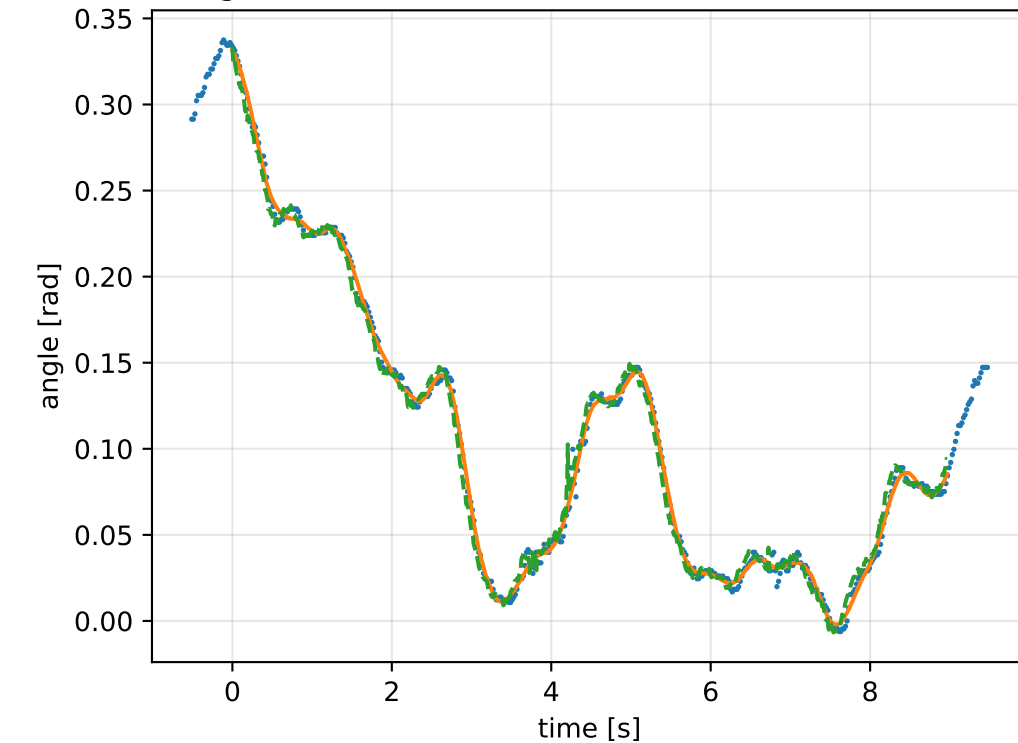
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

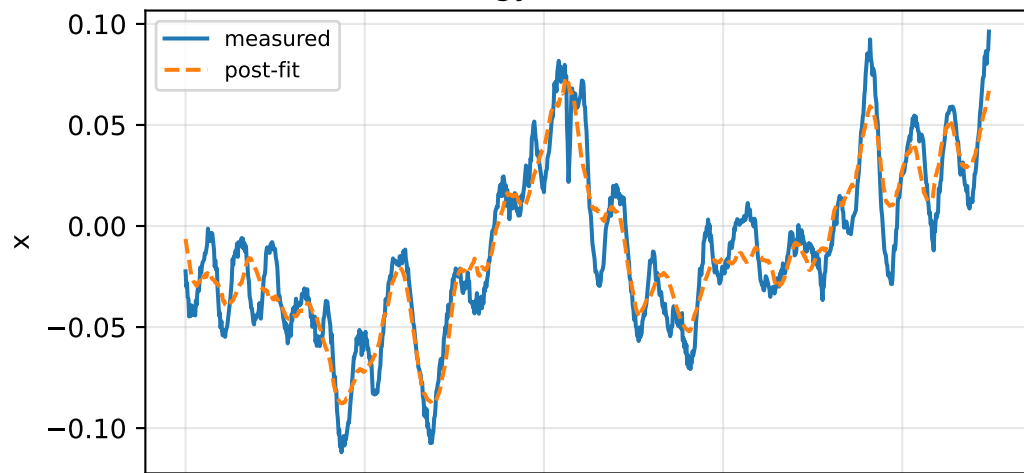


gimbal 4: RMSE=0.006321, max=0.03642 rad

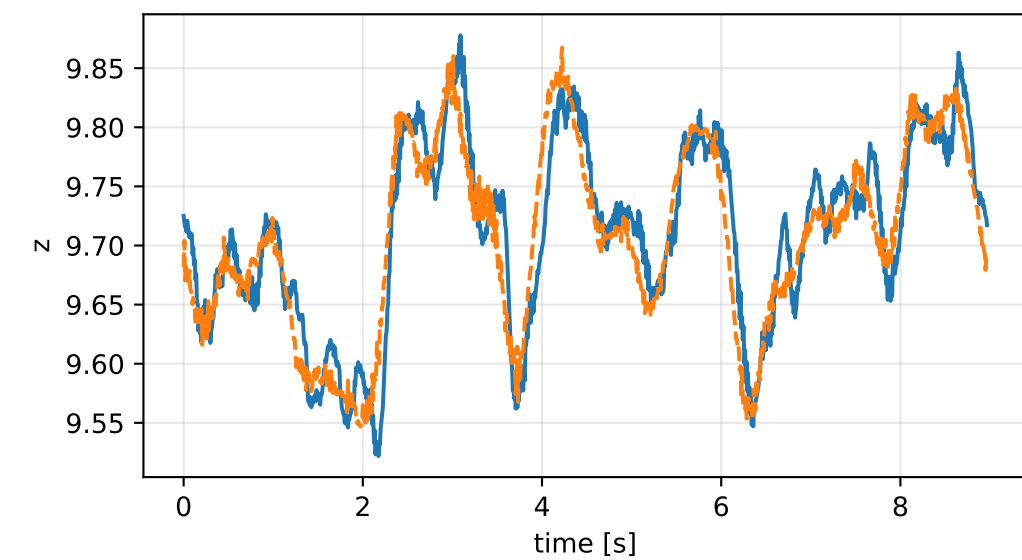
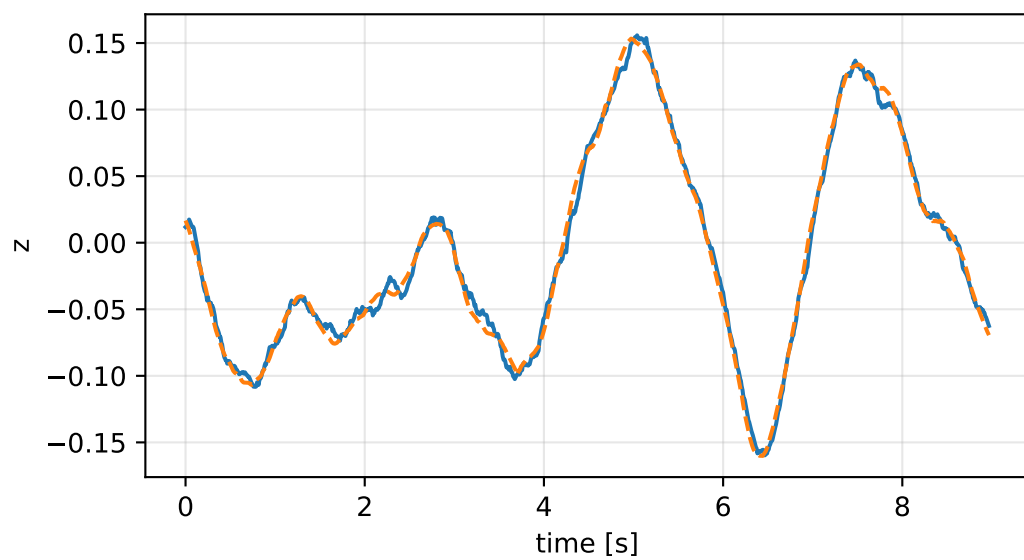
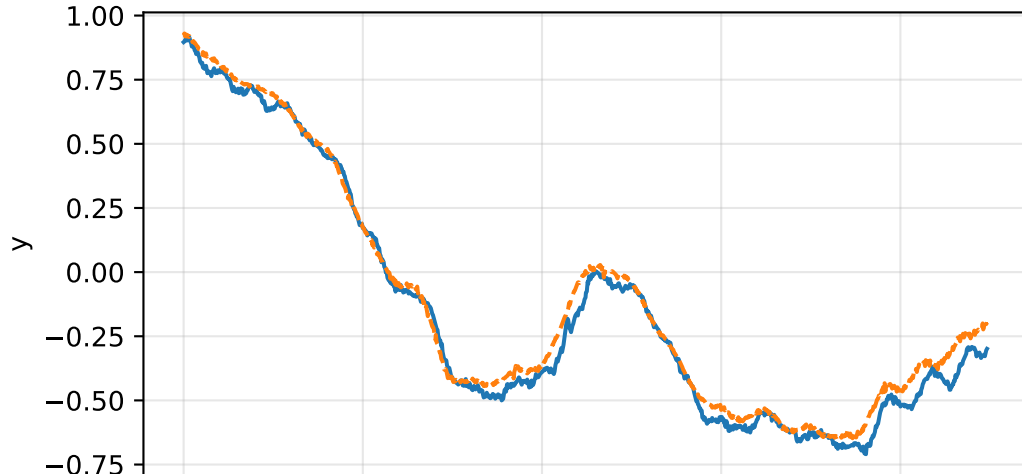
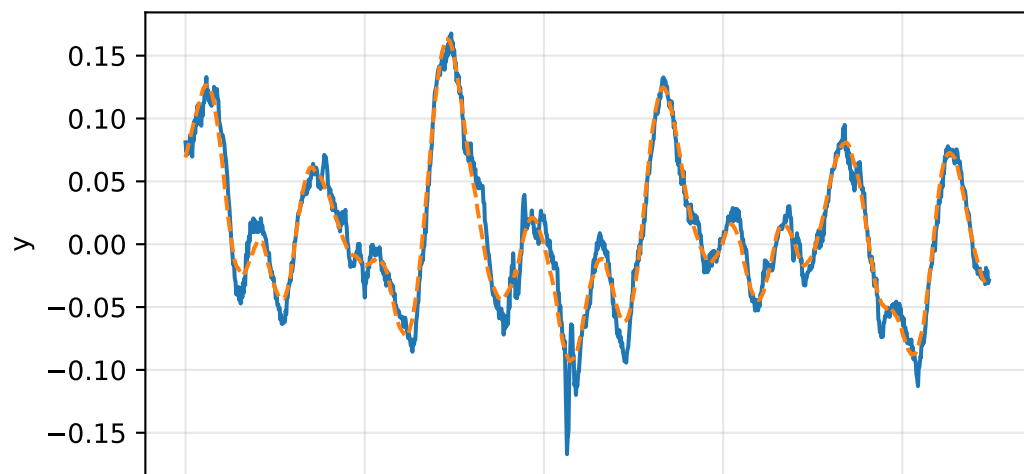
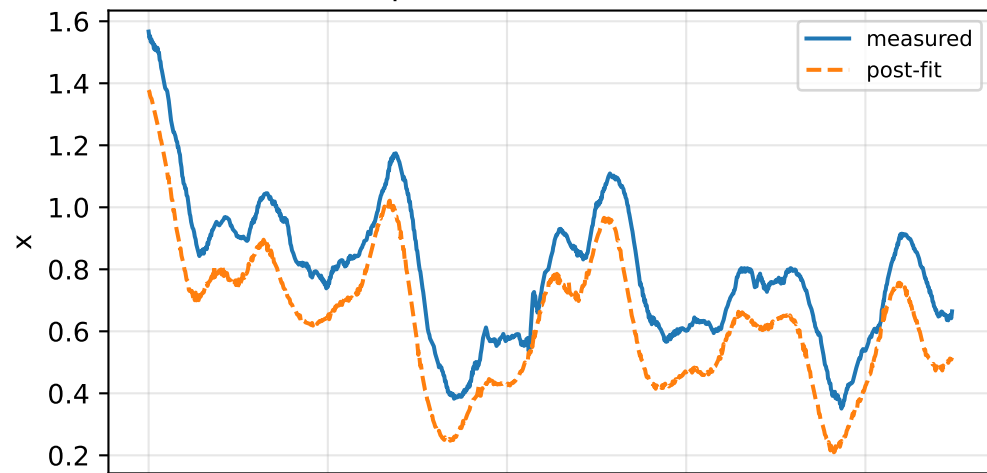


inertia_over_mass_all_nominal: IMU comparison (diagnostic only)

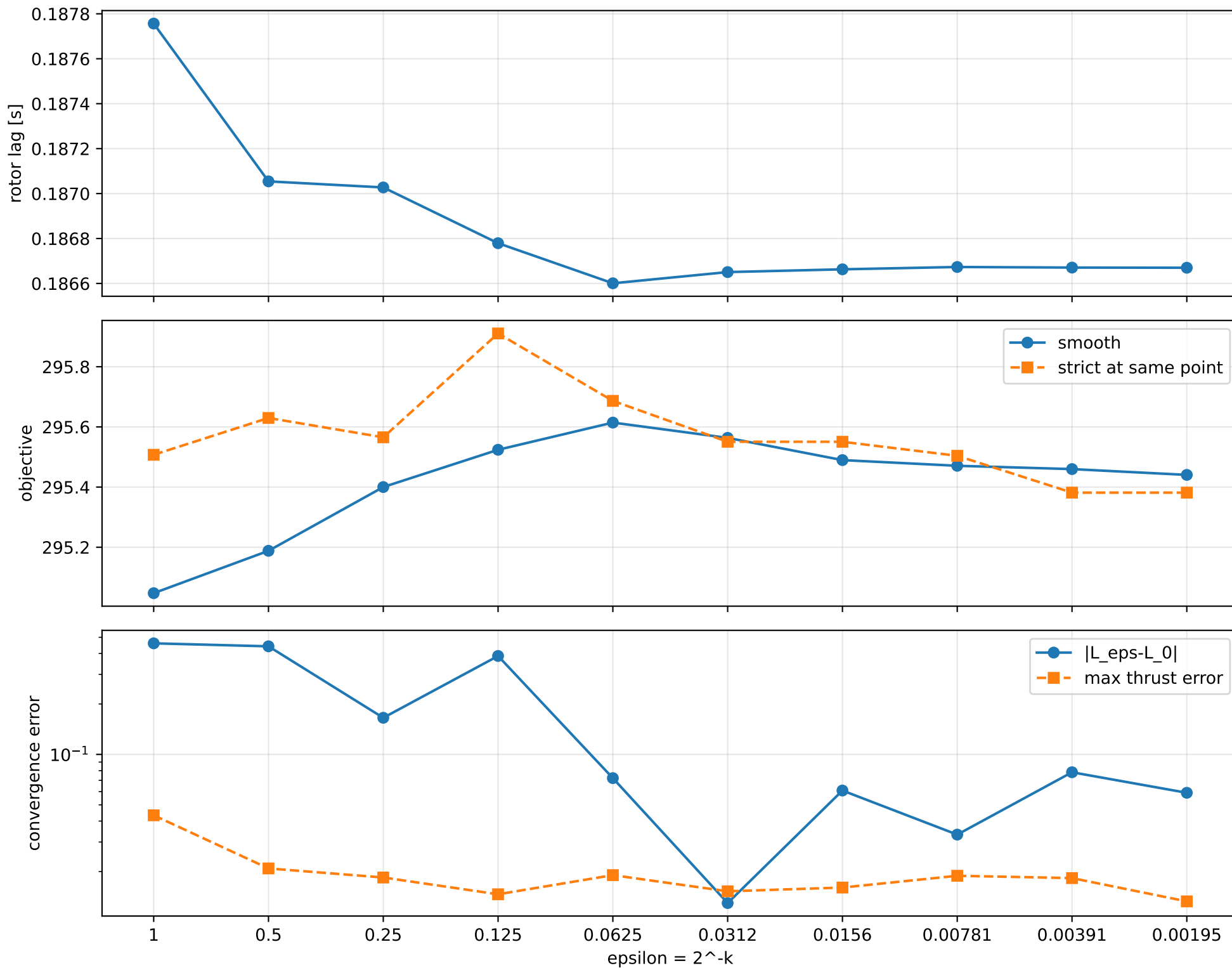
gyro [rad/s]



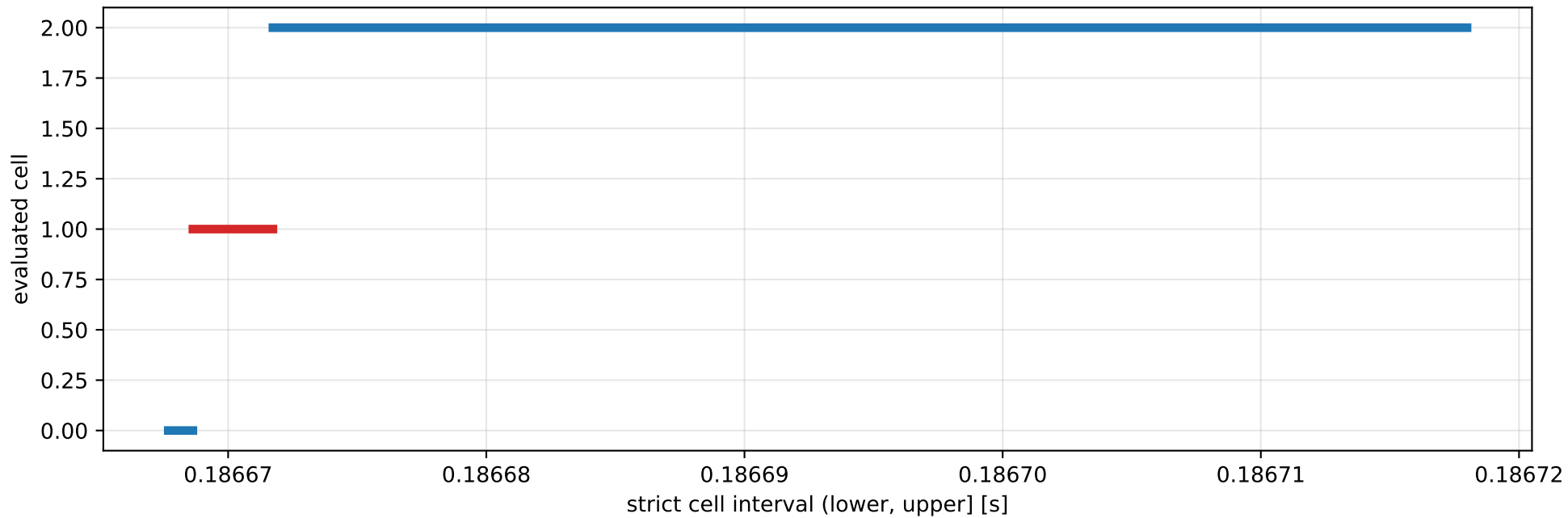
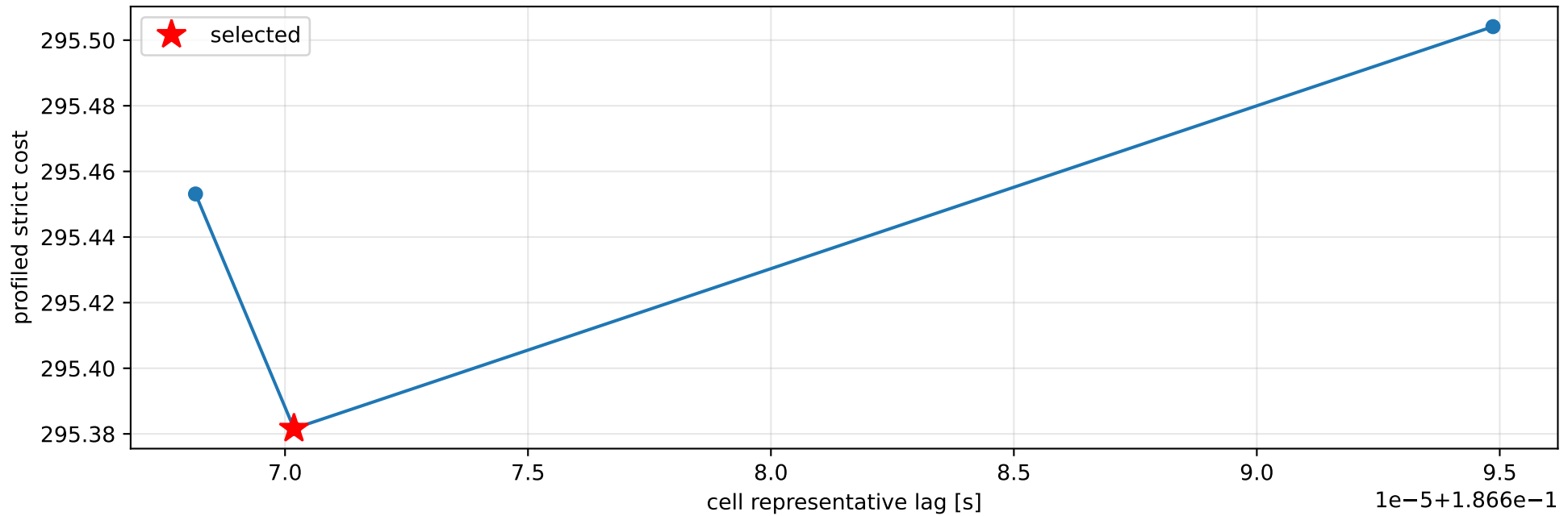
specific force [m/s^2]



inertia_over_mass_all_nominal: smooth-to-strict continuation



inertia_over_mass_all_nominal: exact strict-ZOH lag cells



selected (0.186668634, 0.186671734] s; final smooth 0.18666988 s; data boundary=False

inertia_over_mass_all_nominal: parameter estimate

Case: inertia_over_mass_all_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.67495738

inertia [kg m²]:

[[7.39380073e-02 -8.39323184e-07 2.16288931e-05]

[-8.39323184e-07 7.38841063e-02 7.86765099e-08]

[2.16288931e-05 7.86765099e-08 1.46729357e-01]]

CoG body [m]: [-0.01550974 -0.06094269 0.02759379]

force effectiveness: [1.18134471 1.20592654 0.57735439 0.72614103]

scale-free J/m [m²]:

[[2.76408170e-02 -3.13770676e-07 8.08569635e-06]

[-3.13770676e-07 2.76206667e-02 2.94122480e-08]

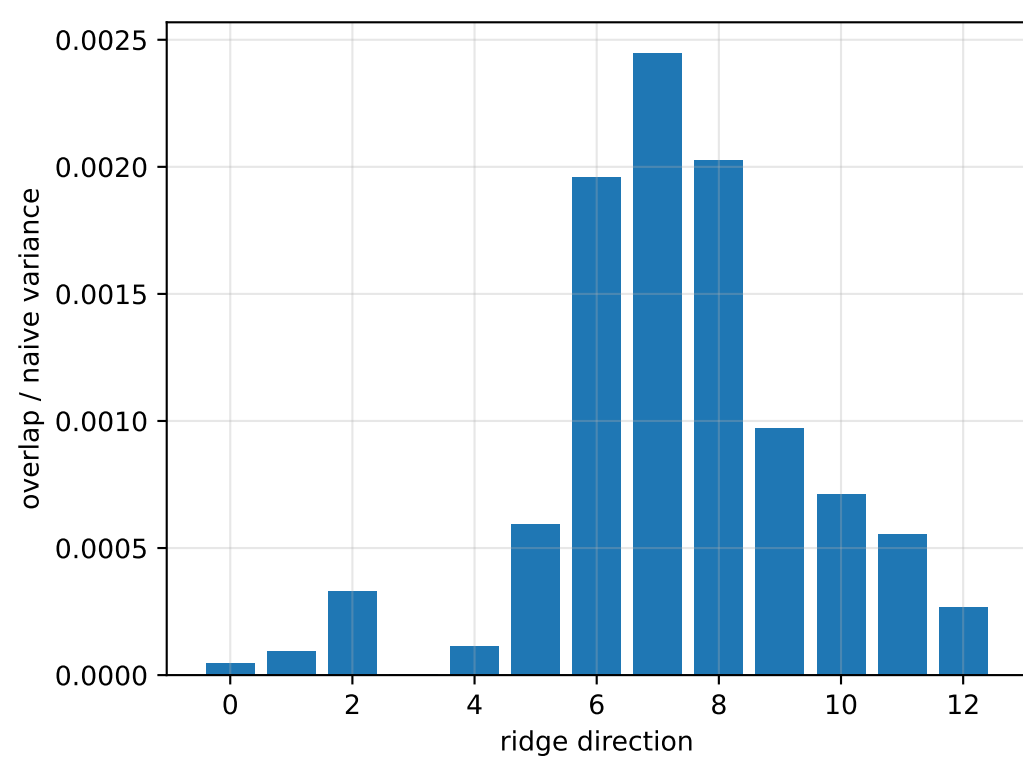
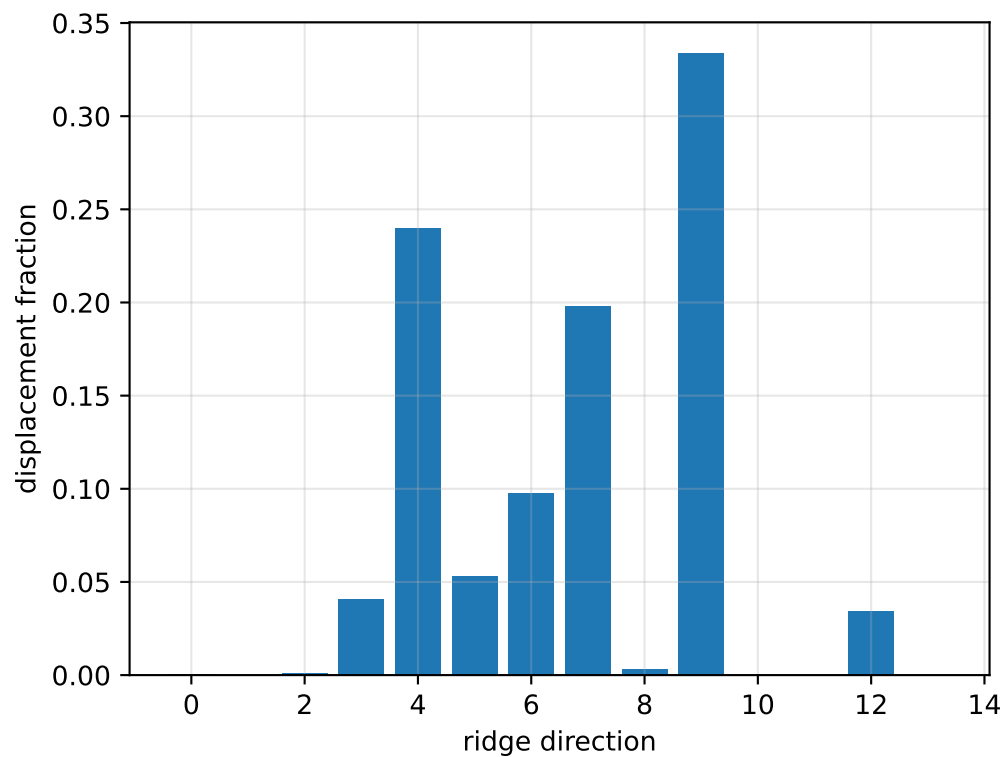
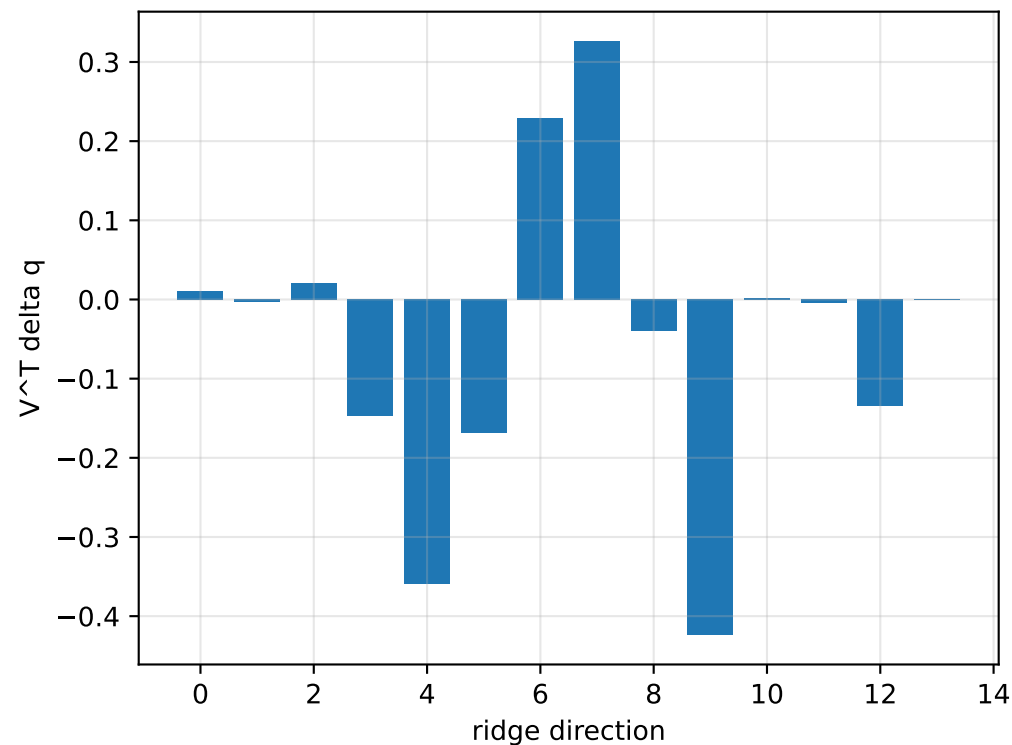
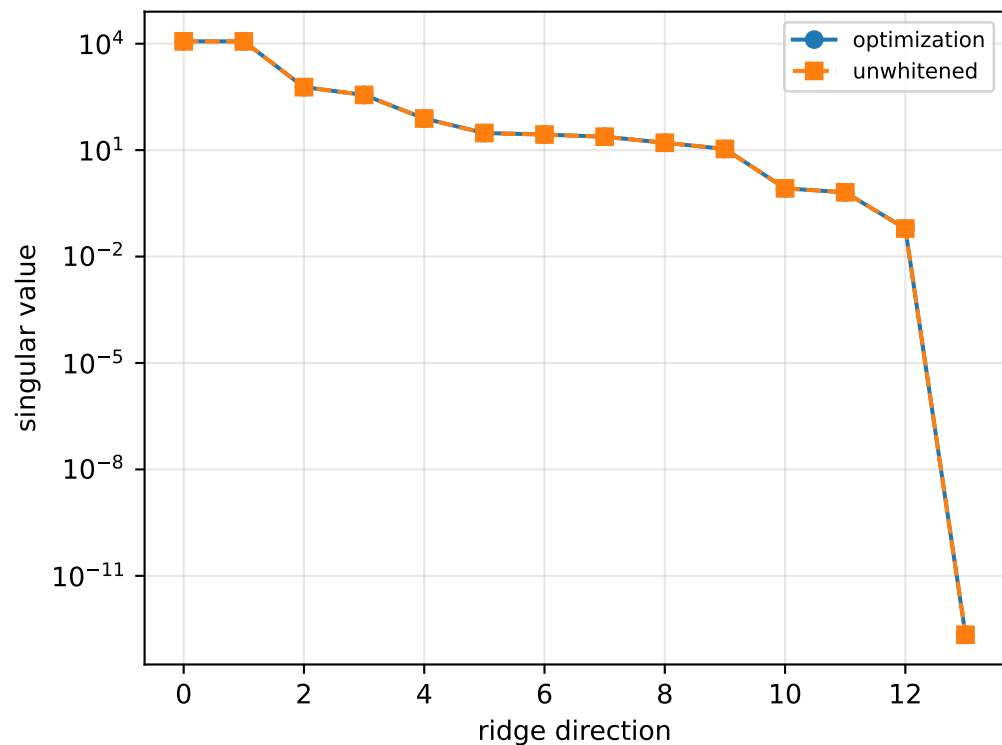
[8.08569635e-06 2.94122480e-08 5.48529702e-02]]

scale-free f/m: [0.44163123 0.45082084 0.21583686 0.27145891]

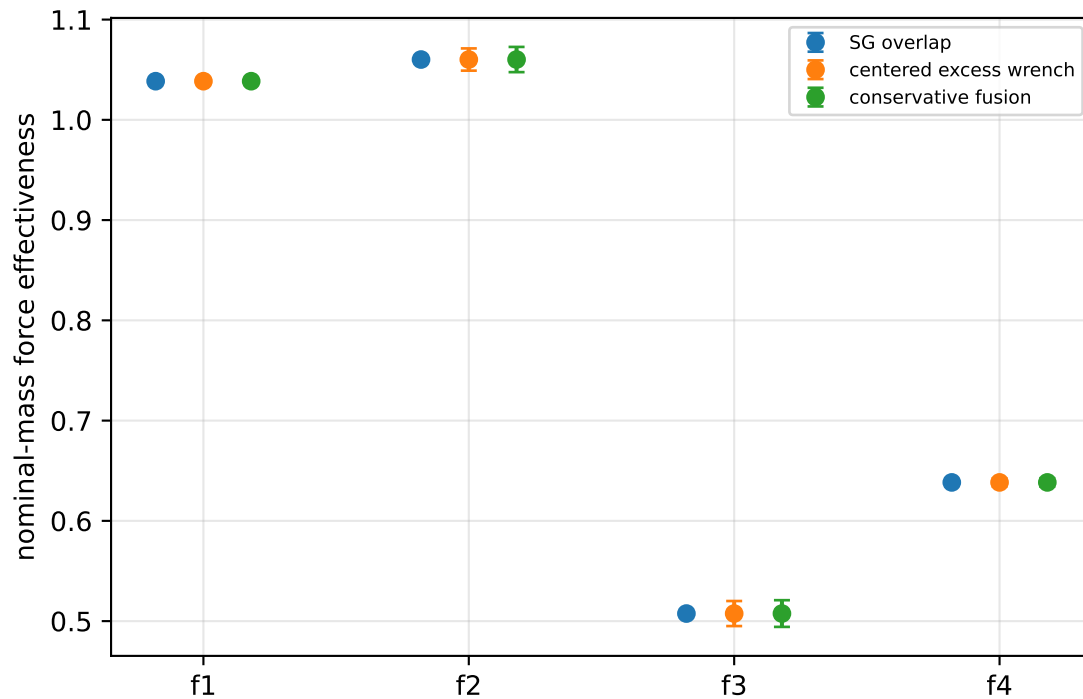
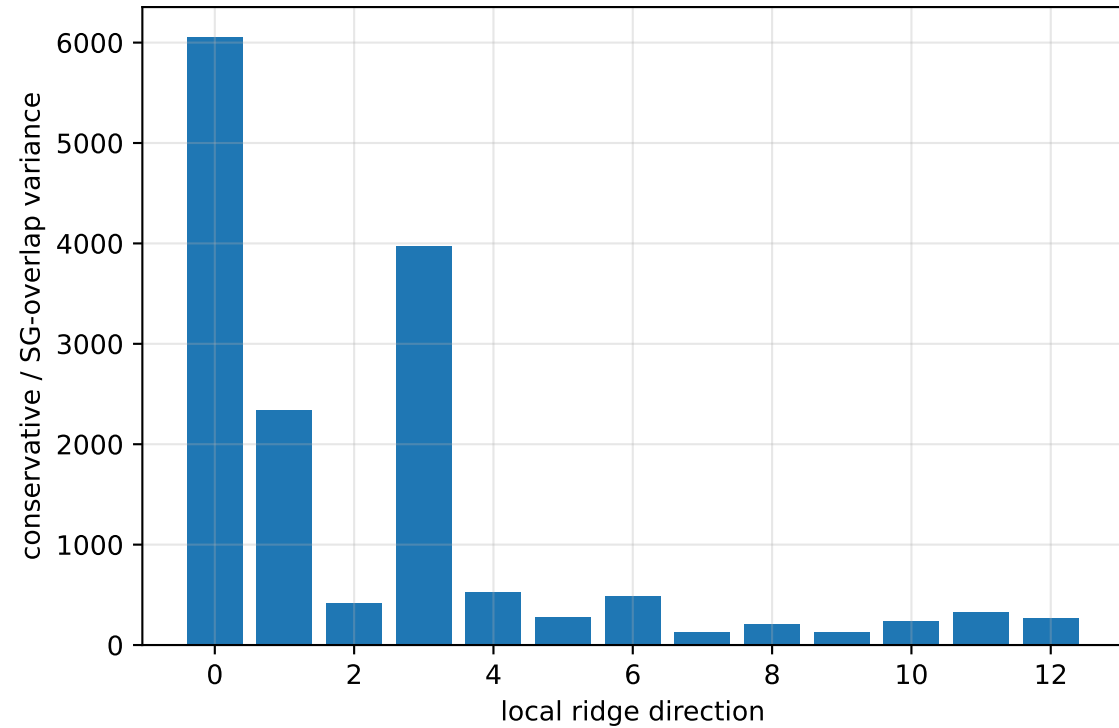
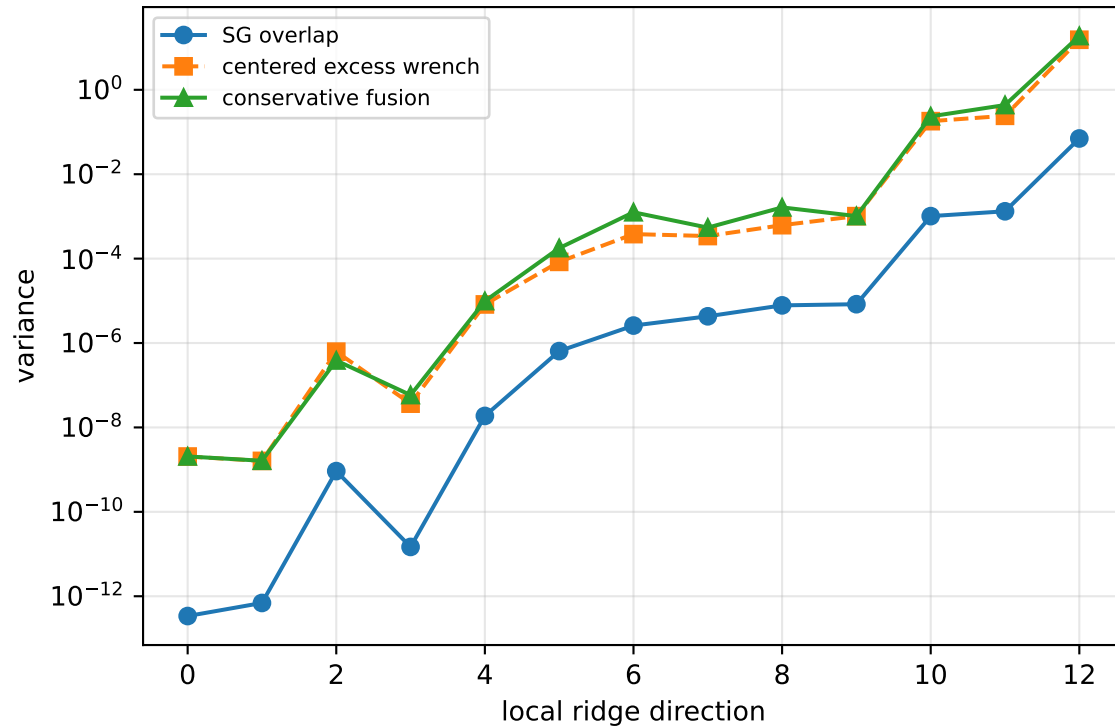
rotor lag cell: (0.186668634, 0.186671734] s

representative: 0.186670184 s

inertia_over_mass_all_nominal: ridge spectrum (rank 13, nullity 1, ||jv||=3.29e-13)



inertia_over_mass_all_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 3550$
wrench-to-acceleration recovery max error = $2.78\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

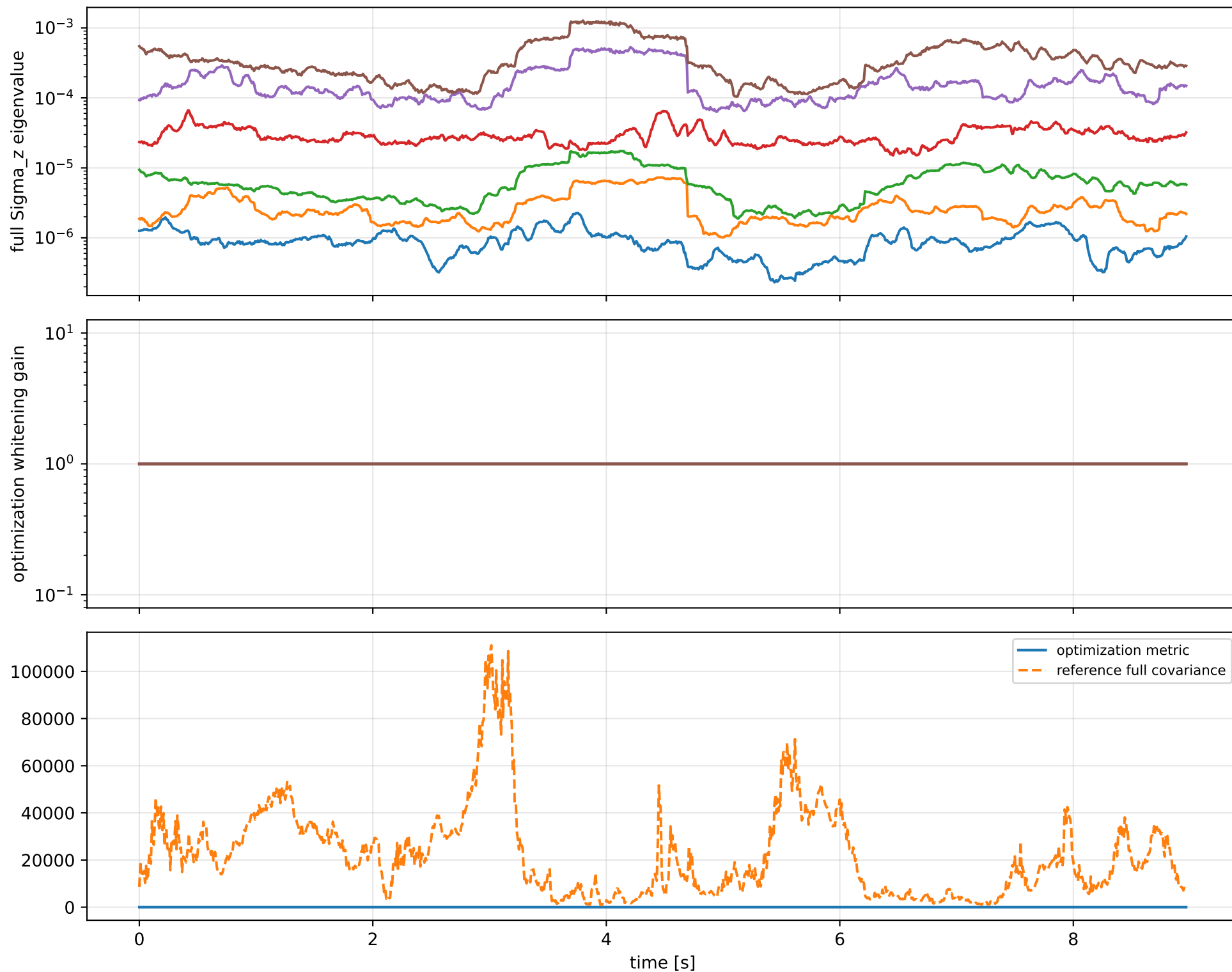
index 12: variance=18.77, ratio=267
 $\log_{\text{mass}}=+0.132$; second_moment_diag_1=+0.139
second_moment_diag_2=+0.138; second_moment_diag_3=-0.935
second_moment_offdiag_12=-0.000231; second_moment_offdiag_13=-0.0049
second_moment_offdiag_23=+0.0284; cog_x=-2.71e-05
cog_y=-4.87e-05; cog_z=-4.8e-06
log_force_eff_1=+0.132; log_force_eff_2=+0.132
log_force_eff_3=+0.131; log_force_eff_4=+0.132

index 11: variance=0.4352, ratio=329.2
 $\log_{\text{mass}}=-0.00171$; second_moment_diag_1=+0.00274
second_moment_diag_2=-0.00138; second_moment_diag_3=+0.00313
second_moment_offdiag_12=-0.00693; second_moment_offdiag_13=+0.959
second_moment_offdiag_23=+0.283; cog_x=+0.000194
cog_y=+0.000457; cog_z=+0.000208
log_force_eff_1=-0.00193; log_force_eff_2=-0.00483
log_force_eff_3=+0.0072; log_force_eff_4=-0.00324

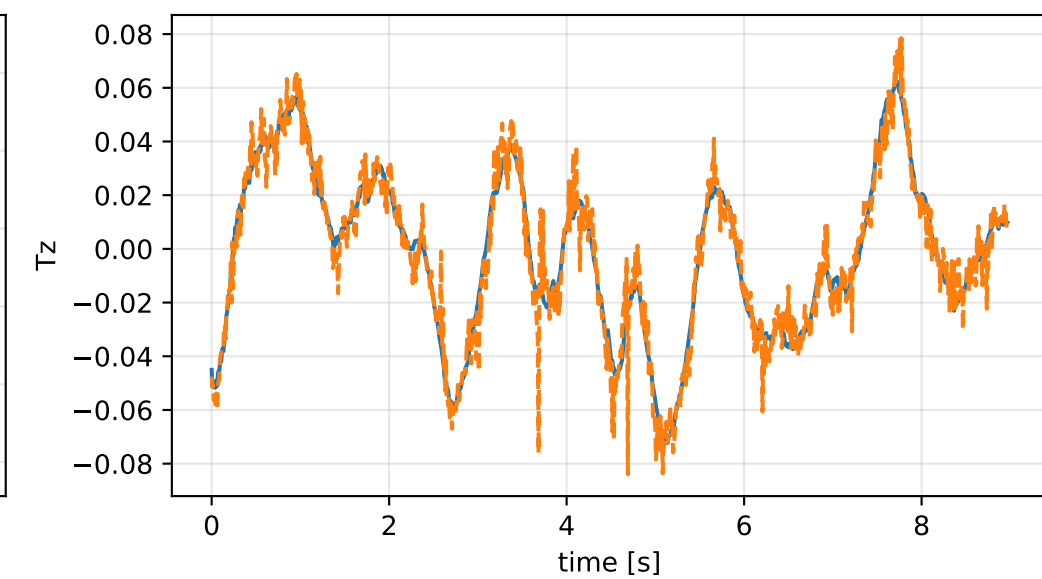
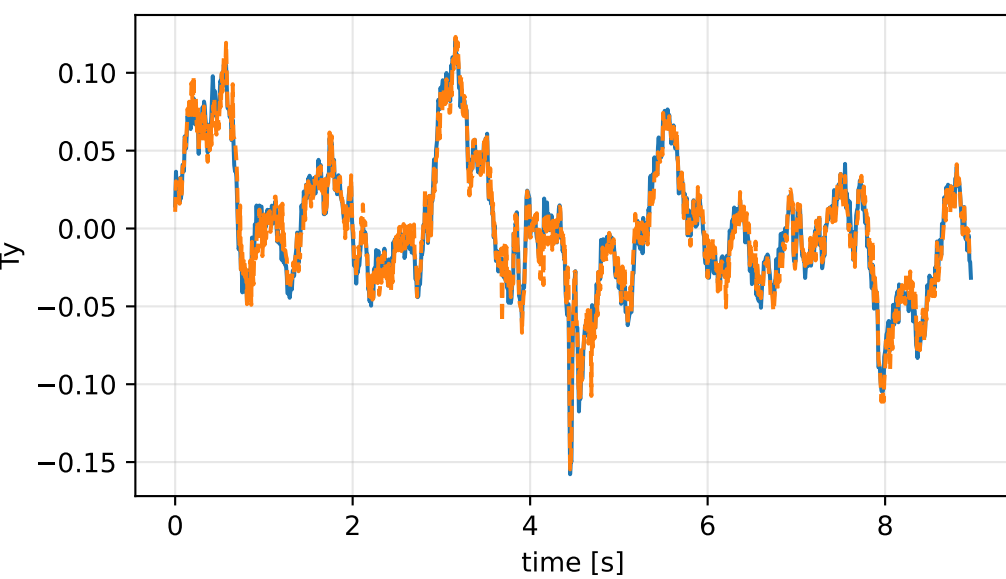
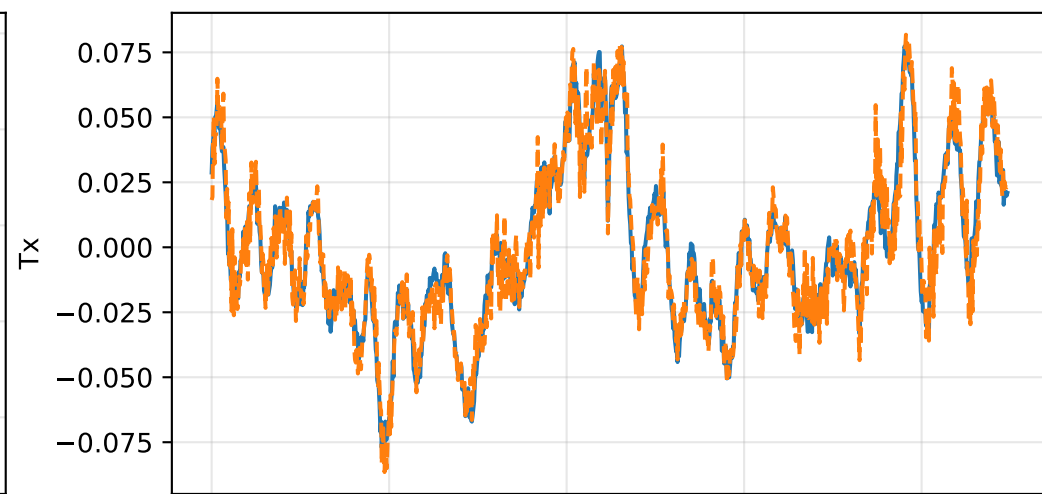
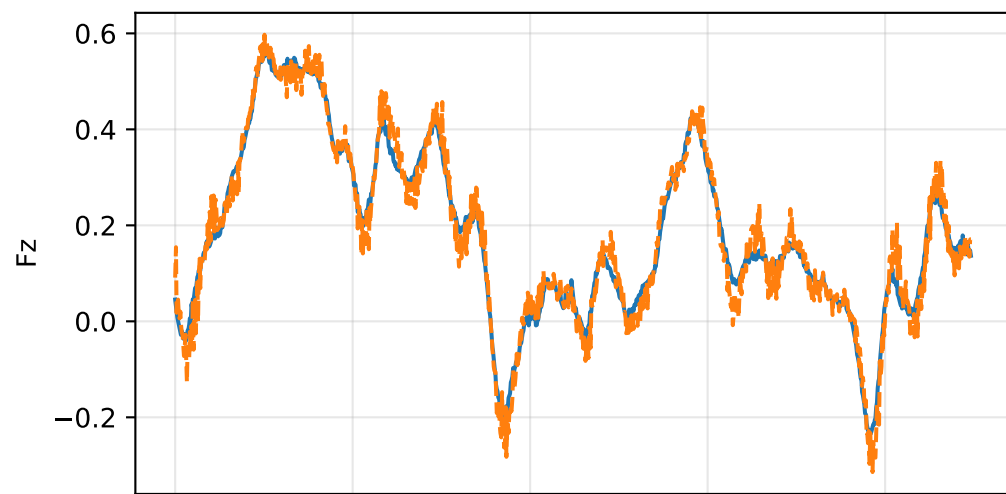
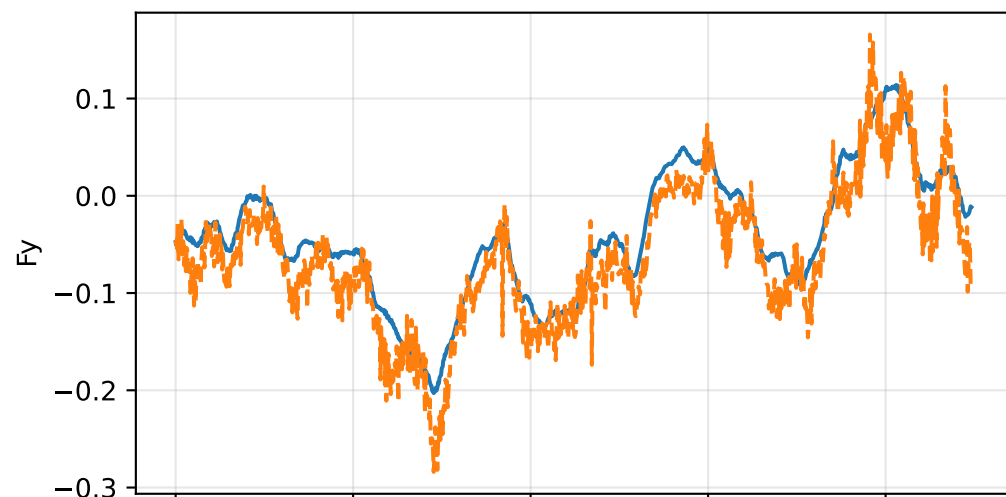
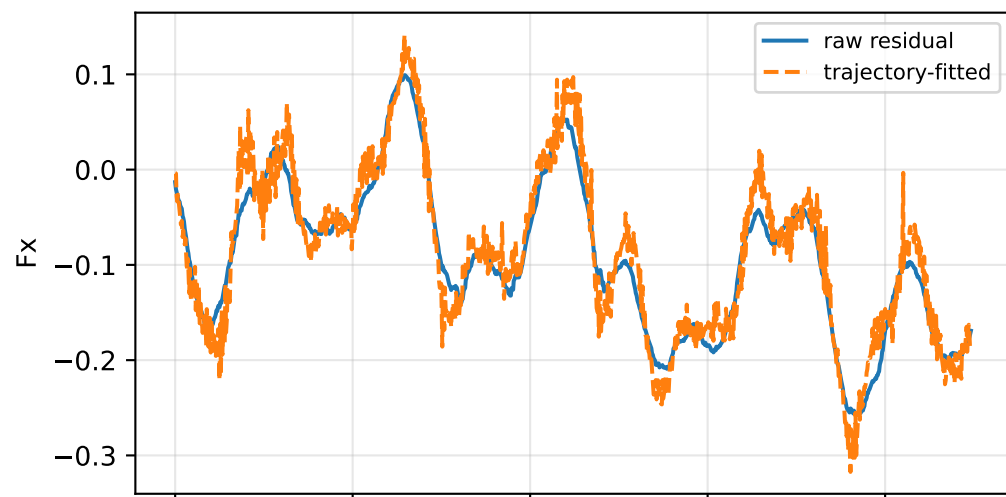
index 10: variance=0.2348, ratio=231.1
 $\log_{\text{mass}}=-0.00734$; second_moment_diag_1=+0.00588
second_moment_diag_2=-0.000498; second_moment_diag_3=+0.0268
second_moment_offdiag_12=+0.00908; second_moment_offdiag_13=-0.283
second_moment_offdiag_23=+0.958; cog_x=+0.000335
cog_y=+0.000464; cog_z=+0.000149
log_force_eff_1=-0.00792; log_force_eff_2=-0.0101
log_force_eff_3=+0.00306; log_force_eff_4=-0.00985

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_all_nominal: optimization vs reference covariance



inertia_over_mass_all_nominal: wrench / closure (force max $7.41\text{e-}15$, torque max $2.78\text{e-}17$)



inertia_over_mass_all_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_all_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/inertia_over_mas

SHA256: 696c77257c9aa3a12bfed1ee5eb82a98ad78c606342ec66cb8ccb4cf17bb8a36

data objective: 295.381461741

prior objective: 0.000168691514424

total objective: 295.381630433

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_all_nominal (inertia_over_mass_m2)

components: xx, yy, zz, xy, xz, yz

target: [2.76408097e-02 2.76206510e-02 5.48529694e-02 -3.09533934e-07
8.08602633e-06 2.51604719e-08]

std: [1.e-06 1.e-06 1.e-06 1.e-06 1.e-06 1.e-06]

final: [2.76408170e-02 2.76206667e-02 5.48529702e-02 -3.13770676e-07
8.08569635e-06 2.94122480e-08]

error: [7.26171661e-09 1.57438191e-08 8.03863746e-10 -4.23674178e-09
-3.29974904e-10 4.25177618e-09]

standardized residual: [0.00726172 0.01574382 0.00080386 -0.00423674 -0.00032997 0.00425178]

factor objective: 0.000168691514424